

1-Page Project Status Report

COMP 2139 – Assignment 2
Virtual Event Ticketing System – Group 145
December 4, 2025

Completed Requirements

1. Enhanced Identity Core

Fully functional ASP.NET Core Identity with registration, login, logout, and role-based authorization. Extended ApplicationUser model includes FirstName, LastName, and timestamps. Three roles (Admin, Organizer, Attendee) created and seeded. Email confirmation configured with RequireConfirmedEmail setting.

2. My Dashboard

My Tickets section displays user tickets with QR codes, PDF download functionality using iText7, and 1-5 star rating system for events. Dashboard accessible at /Dashboard/MyTickets with complete ticket management.

3. AJAX Live Search and Filtering

Real-time event search with 300ms debounce, instant category filtering, and dynamic sorting using _EventPartial.cshtml. No page reloads required, loading spinner during searches, and low-stock warnings display. Implemented with jQuery AJAX and partial view rendering.

4. AJAX Shopping Cart with Live Counter

Session-based cart with live counter badge in navbar. AJAX endpoints for add/update/remove operations with real-time updates. Low-stock warnings when fewer than 10 tickets available. Cart persistence across pages using JSON serialization.

5. AJAX Purchase Confirmation Modal

Bootstrap modal with canvas-confetti animation library. Success message with order summary, animated confetti burst effect, ticket download button in modal, and automatic cart clearing after purchase.

6. Global Error Handling

Custom 404/500 error pages with user-friendly messages. Serilog logging configured with console sink, enrichers (MachineName, Application), and comprehensive logging for login, registration, purchases, and exceptions.

7. Production Configuration

appsettings.Production.json configured with secure cookies (HttpOnly, SecurePolicy: Always), HTTPS enforcement, CSP headers, and DataProtection registration. Application ready for Azure deployment.

8. Unit Testing

xUnit test framework with in-memory database. 14+ tests covering model validation, event/purchase/cart controllers, and AJAX endpoints. Tests validate purchase flow, event filtering, cart operations, and low-stock warnings.

9. Forgot/Reset Password UI Flow

Complete password reset workflow with ForgotPasswordViewModel and ResetPasswordViewModel. Four dedicated views: ForgotPassword page with email input, ForgotPasswordConfirmation page showing email sent message, ResetPassword page with password strength indicator and token validation, and ResetPasswordConfirmation success page. Professional HTML email template with responsive design and cross-client compatibility. Token lifespan configured to 24 hours with DataProtectionTokenProviderOptions. Security features include no user enumeration, secure token generation, and email verification requirement. Integrated with existing EmailSender service supporting both SMTP delivery and development fallback logging.

10. My Events Dashboard Section

Organizer dashboard at /Dashboard/MyEvents with role-based authorization for Organizers and Admins. MyEventsViewModel aggregates event data with ticket sales and revenue using optimized LINQ queries. Dashboard displays summary cards showing total events, tickets sold, and revenue. Detailed table shows per-event statistics including available tickets, tickets sold, and revenue with color-coded stock warnings. Quick stats section provides average revenue per event, average tickets sold, and upcoming events count. Edit and delete action links for event management. Authorization checks ensure organizers only access their own events via OrganizerId filtering.

11. Purchase History Section

Purchase history page at /Dashboard/PurchaseHistory with comprehensive filtering system. PurchaseHistoryFilterViewModel supports date range filtering with UTC to local time conversion, category filtering, and past events toggle. Summary cards display total tickets purchased, total spent, and past events count. Table shows event details, purchase dates, prices, ratings with star display, and event status badges. Filter form includes start date, end date, category dropdown, and past events checkbox. Integration with existing rating system from My Tickets. Clean filters button resets all parameters. Proper timezone handling ensures accurate date comparisons across different time zones.

Uncompleted Requirements

1. Organizer Analytics Dashboard

Not implemented. Requires /Events/MyAnalytics endpoint with Chart.js visualizations including ticket sales by category chart, revenue per month chart, and Top 5 best-selling events table with JSON endpoint. **Reason:** Complex data aggregation required across multiple database tables (Events, Tickets, Purchases) with proper date filtering and grouping. Chart.js library integration needed significant research for proper data formatting and real-time updates. PostgreSQL query optimization for revenue calculations across large datasets presented performance challenges that required additional time for proper implementation.

2. Profile Management Page

Not implemented. User data stored in ApplicationUser model but no edit interface exists. Should allow editing name, email, phone, uploading profile picture to wwwroot/uploads/, and password changes. **Reason:** File upload functionality required implementing secure file validation (file type, size limits), storage management, and serving uploaded images with proper caching headers. Changing email addresses required re-verification workflow to prevent account hijacking. Password change needed additional security considerations including current password validation and session management.